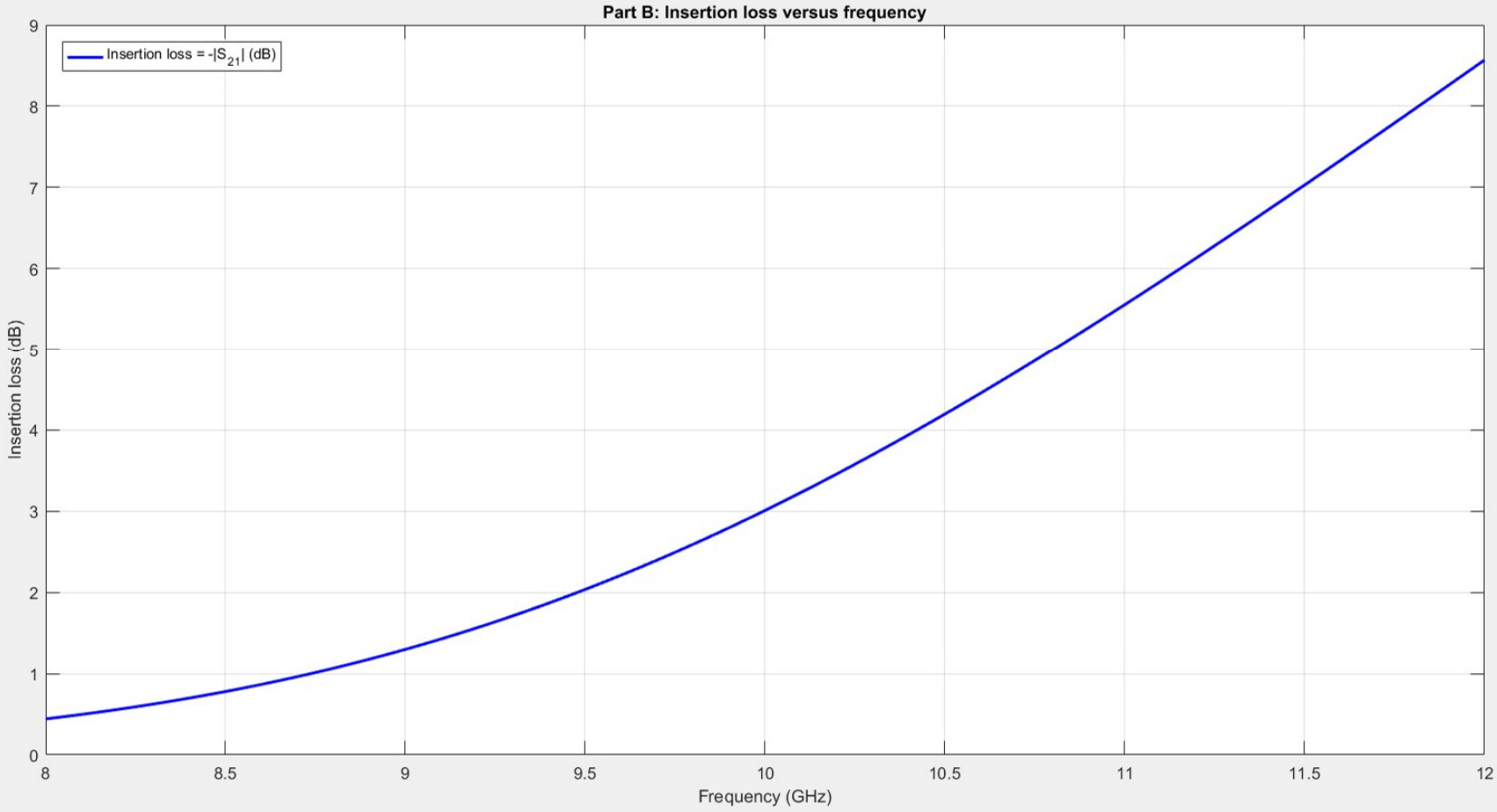
Figure 6

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Exp05.m

```
11 -    fprintf('==================== PART A: IMP
12
13 -    % ---- Given data ----
14 -    Z0 = 50;                    % characteris
15 -    f0 = 10e9;                  % design freq
16 -    ZL = 75 + 25j;              % load impeda
17 -    c  = 3e8;                   % speed of li
18 -    lambda0 = c/f0;             % wavelength
19
20 -    f = linspace(8e9, 12e9, 801);   % simulat
21 -    N = length(f);
22 -    [~, i0] = min(abs(f - f0));     % index o
23
24 -    % ---- 1 & 2. Theoretical load reflection
25 -    GammaL      = (ZL - Z0)/(ZL + Z0);
26 -    GammaL_mag  = abs(GammaL);
27 -    GammaL_ang  = angle(GammaL)*180/pi;
28 -    RL_theory   = -20*log10(GammaL_mag);
29 -    VSWR_load   = (1 + GammaL_mag)/(1 - Gamma
30 -    MismatchLoss = -10*log10(1 - GammaL_mag^2
31
32 -    fprintf('\n--- Load (before matching), th
33 -    fprintf('Load impedance ZL          = %.2f
34 -    fprintf('Reflection coeff. GammaL   = %.4f
35 -    fprintf('|GammaL|                   = %.4f
36 -    fprintf('Angle of GammaL            = %.2f
37 -    fprintf('Initial return loss        = %.3f
38 -    fprintf('VSWR                       = %.4f
39 -    fprintf('Mismatch loss              = %.4f
40
```

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```
==================== PART A: IMPEDANCE MATCHING ====================

--- Load (before matching), theoretical ---
Load impedance ZL          = 75.00 + j25.00 ohm
Reflection coeff. GammaL   = 0.2308 + j0.1538
|GammaL|                   = 0.2774
Angle of GammaL            = 33.69 deg
Initial return loss        = 11.139 dB
VSWR                       = 1.7676
Mismatch loss              = 0.3476 dB

--- Single-stub design solutions (open-circuit shunt stub) ---
Solution 1: d = 0.1942 lambda (5.825 mm), l = 0.4167 lambda (12.500 mm), y(d) = 1.0000 +0.5774j
Solution 2: d = 0.3994 lambda (11.983 mm), l = 0.0833 lambda (2.500 mm), y(d) = 1.0000 -0.5774j
Selected solution 2 (shortest total length):
   Series line length d = 0.3994 lambda = 11.983 mm
   Open stub length   l = 0.0833 lambda = 2.500 mm

--- Matched input impedance (theoretical, 10 GHz) ---
Zin = 50.0000 +0.0000j ohm,  |Gamma_in| = 1.42e-16

--- Comparison at 10 GHz: theoretical vs simulated ---
Quantity                      Theoretical      Simulated
|Gamma| before matching       0.2774           0.2774
S11 before (dB)               -11.139          -11.139
Return loss before (dB)       11.139           11.139
Re{Zin} after (ohm)           50.0000          50.0000
Im{Zin} after (ohm)           0.0000           0.0000
|Gamma| after matching        1.42e-16         8.60e-17
Return loss after (dB)        240.0            240.0
(Values near 1e-7 or below are numerical round-off; ideal value is 0.)
```

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38 -    fprintf('VSWR                       = %.4f
39 -    fprintf('Mismatch loss              = %.4f
40
```

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```
(values near 1e-7 or below are numerical round-off, ideal value is 0.)

Matched bandwidth (RL >= 10 dB): 8.000 GHz to 11.880 GHz  (3.880 GHz, 38.8 % fractional)
(If the lower edge equals 8 GHz, it is limited by the sweep range.)
Worst-case RL after matching over 8-12 GHz = 9.51 dB

==================== PART B: FILTER DESIGN ====================

Filter: 5-order Butterworth low-pass, fc = 10.0 GHz, Z0 = 50 ohm
Prototype g-values: 0.6180 1.6180 2.0000 1.6180 0.6180
  Element 1: shunt C1 = 0.1967 pF
  Element 2: series L2 = 1.2876 nH
  Element 3: shunt C3 = 0.6366 pF
  Element 4: series L4 = 1.2876 nH
  Element 5: shunt C5 = 0.1967 pF

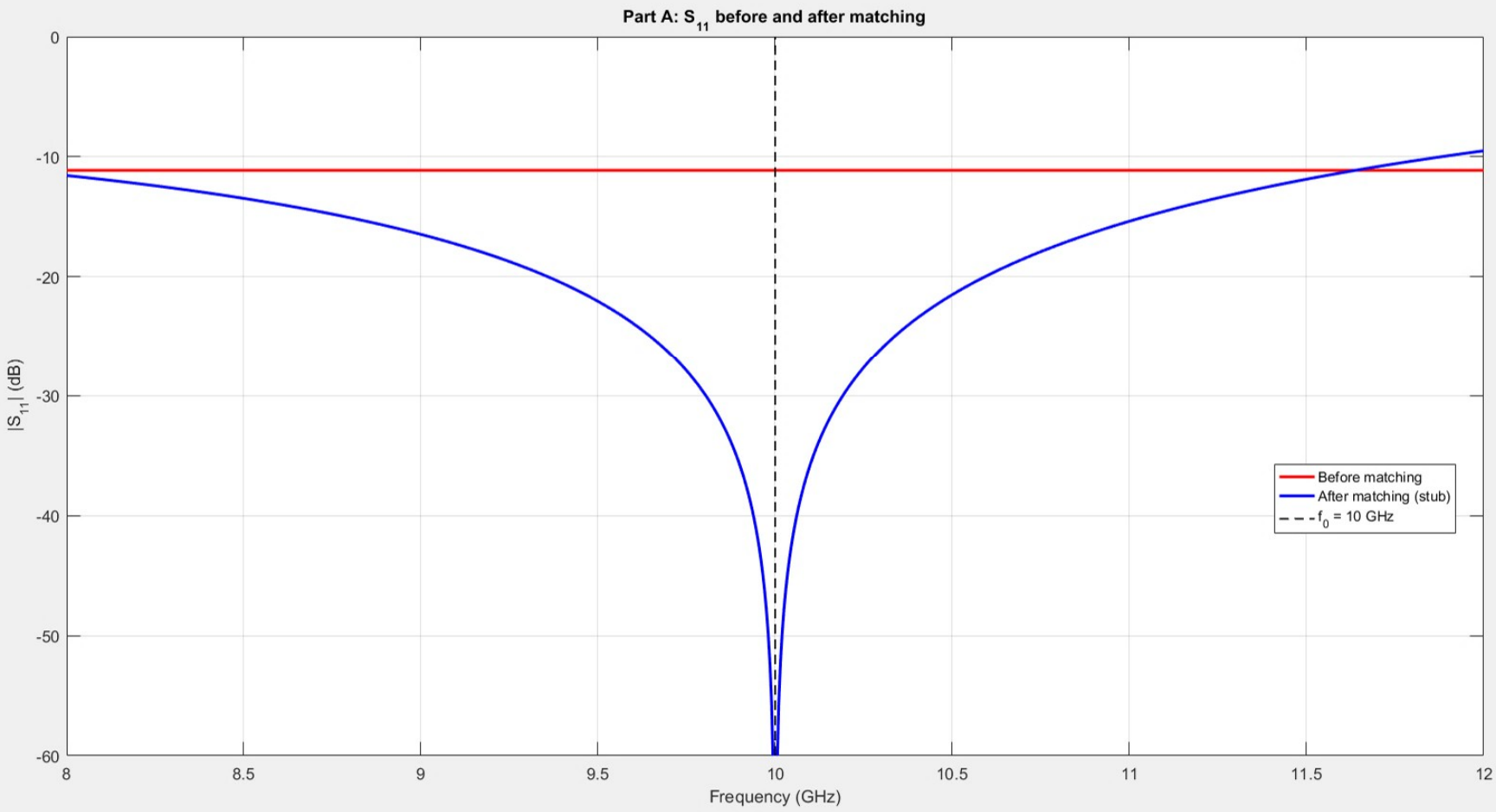
--- Filter results ---
Simulated -3 dB cutoff frequency = 10.000000 GHz
Specified cutoff frequency       = 10.000000 GHz
Percentage deviation             = 0.000e+00 %
Passband  : 8.000 GHz to 10.000000 GHz
Stopband  : 10.000000 GHz to 12.000 GHz
Insertion loss at 8 GHz          = 0.443 dB
Average insertion loss (passband)= 1.444 dB
Maximum insertion loss (passband)= 3.010 dB (at cutoff edge)
S21 at 12 GHz (stopband)         = -8.568 dB  (attenuation 8.568 dB)
S11 at 8 GHz                     = -10.134 dB
S11 at 12 GHz                    = -0.650 dB
Max |S21| difference, simulated vs theoretical Butterworth = 1.155e-14 dB

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Figure 1

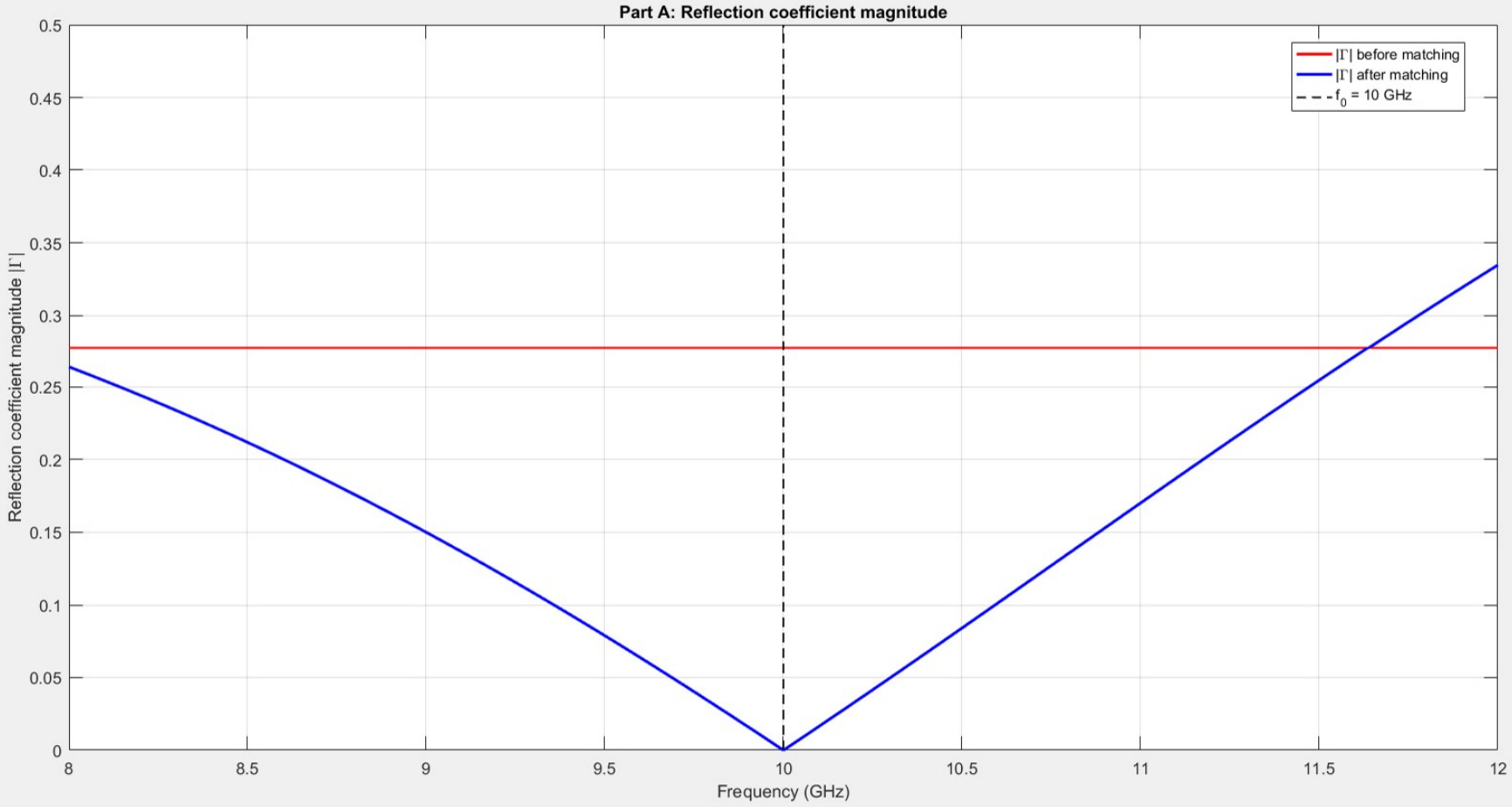
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Figure 2

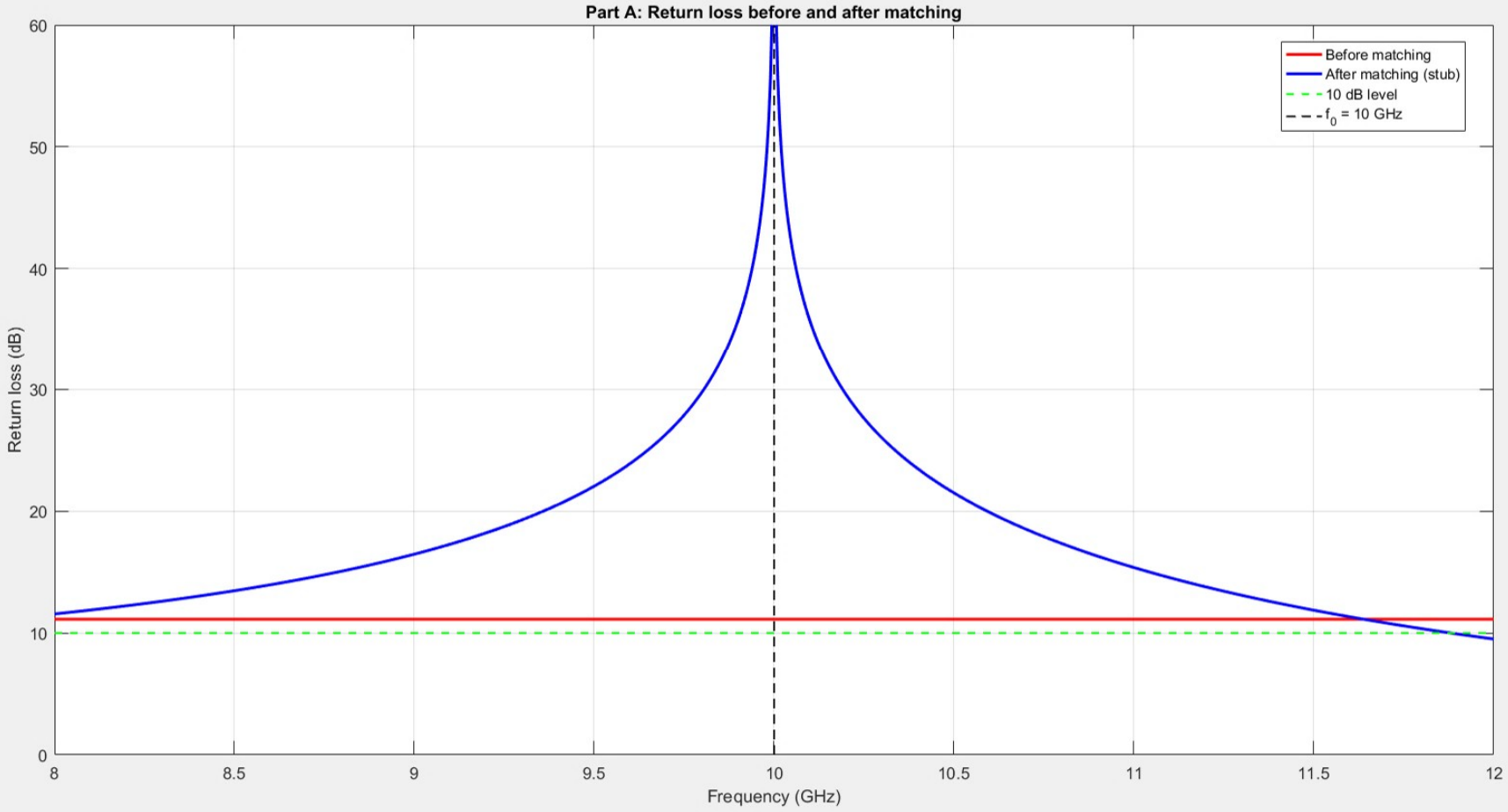
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Figure 3

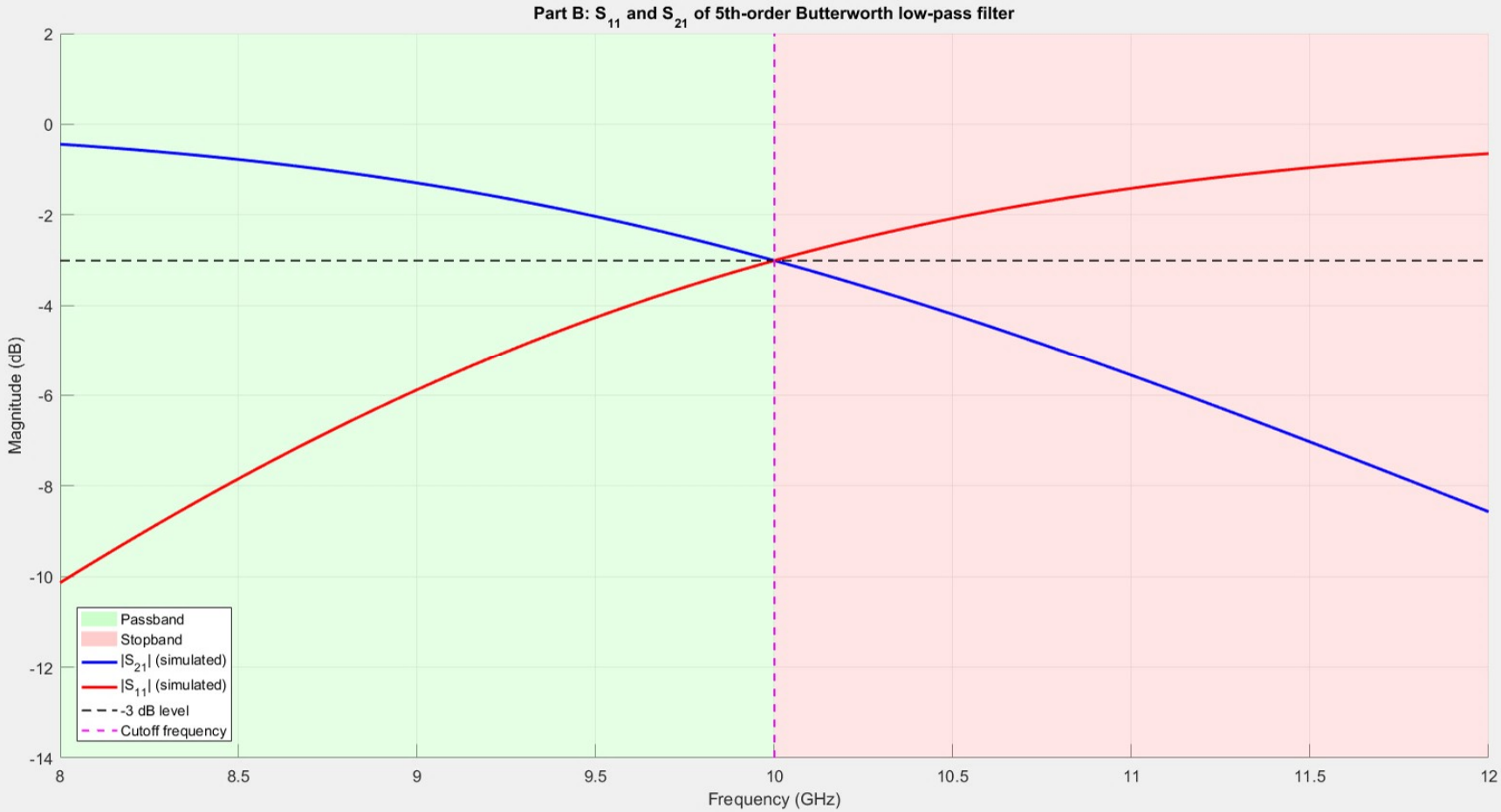
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Figure 4

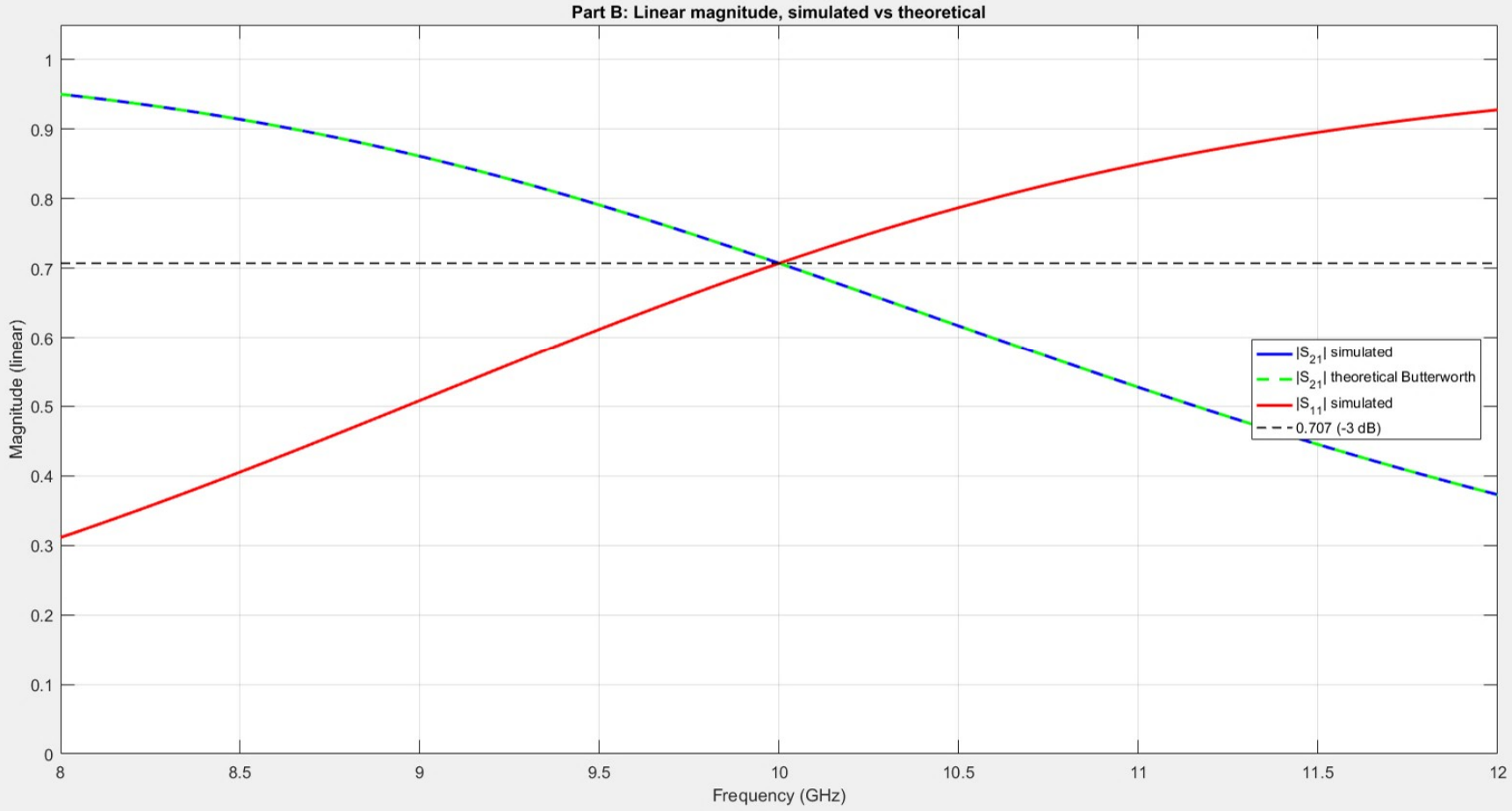
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